# 2 Open Raise

03/10/2019

# Open Manual Raise Applicability I

*Raise is of generally short length.

*Raise is of small cross section Maximum 2m X 2m.

*Raise has to take turn in between.

*Raise can to follow ore body.

*Raise with changing cross section.

# Open Manual Raise Process I

*Manual raise up to 70 degree inclination drilled by Jack Hammer with pusher air leg.

*Vertical and steeply Deeping raise are drilled with Stopper.

*Area, centerline are marked at place, where raise has to be excavated.

*Center line for raise is fixed considering both, starting place at lower level and point of puncher at upper level.

*First blast for raise is taken from floor of drive / cross cut.

# Open Manual Raise Process II

*Next few blasts are taken by standing over accumulated muck pile.

*3 to 4 meter raising is possible with out erecting platform in raise.

*Normally 1.2m round is drilled at face for blasting

*This process continues, till drill men are able to reach raise face standing over broken rock.

*Drilling starts with 0.6m integrated  drill steel rod.

# Open Manual Raise Process III

*Manual open raise can be up to 100 m long.

*For 100 m raise, two raises were taken parallel, with connection at 30m interval.

*One raise shall be equipped with wooden ladder etc, while other raise at progress mode rope ladder.

*When raise angle is less than angel of repose, scrapper is deployed for scrapping broken rock down along raise.

*For raise using scrapper, climbing is done with help of rope in place of rope ladder.

# Open Manual Raise Process IV

*At last blast, four no of clamp holes are drilled at both side of raise, two at each side.

*Wooden / steel net platform is erected with help of clamp holes.

*One steel rope ladder is placed on floor of raise of sufficient length.

*All accumulated muck is removed with loader at lower level.

*Subsequent drilling, blasting is done with help of platform erected in raise.

# Open Manual Raise Process V

*Normally, raise has tendency to become flat.

*To mitigate this tendency, drill hole direction is kept at 10 degree higher than raise angle (inclination).

* This also prevents damage of compressed air and water pipe fitted at raise.

# Open Manual Raise Survey I

*Centre line is placed at back (roof) of drive.

*At least two center holes are drilled along center line.

*Wooden Peg along with pin are placed in the holes.

*Line of sight from two threads shall provide centre line of raise.

*Similarly, grade lines pages are paced at both side of raise for extending the grade line.

*With increasing distance, accuracy of center line and grade line both gets reduced.

*Both center line peg and grade pegs are shifted in side raise with help of Theodolite.

# Open Manual Raise Survey II

*Raise is surveyed with Theodolite after 10 to 15 m progress.

*Necessary correction in directions are made after survey.

*At point, where angle or direction of raise has to change, new centre line and grade lines are provided.

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# Open Manual Raise Ventilation I

*Ventilation is maintained through 25mm dia two GI pipe line paced at upper corner of raise with help of compressed air.

*Immediately after blasting, compressed air and water is released at raise face.

*After ventilating face, one GI pipe is connected to compressed air and another to water line for face washing, drilling etc.

*When parting between raise face and upper level becomes 10 to 15 m, a holes is drilled for ventilation of raise & direction of raise.

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# Open Manual Raise Blasting I

*Blaster charge explosive to round drilled at face from platform.

*Platform then dismantled and placed in man hole, excavated at side wall of raise.

*Man hole are excavated at interval of 8 to 10 meter along raise.

*Firing blast round is done from safe palce at level.

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# Open Manual Raise Dressing I

*Support man plays a very important role at manual open raise.

*Support man (Timber men) has to get  in to freshly blasted raise.

*They ventilate and spray blasted face with water and air.

*Support man has to look for misfire etc at face, and report to blaster as required.

*Minor damage, displacement of different fittings due to blasting is attended by support men.

# Open Manual Raise Dressing II

*Support men erect working platform below raise face by inserting rail clamp in side hole drilled for this purpose.

*Platform is made of either wooden plank 50mm thick & 200 mm wide or strong wire net (12mm dia, 50mm X 50mm gape).

# Open Manual Raise Advantage I

*Open manual raise requires least preparation.

*Excavation of raise can start very fast.

*Raising has lot of  flexibility to suite requirement.

*Process of raising is very simple.

*Requirement of equipment for raining is minimum.

*Raise can follow ore body, providing prior information of ore and ore body.

# Open Manual Raise Disadvantage I

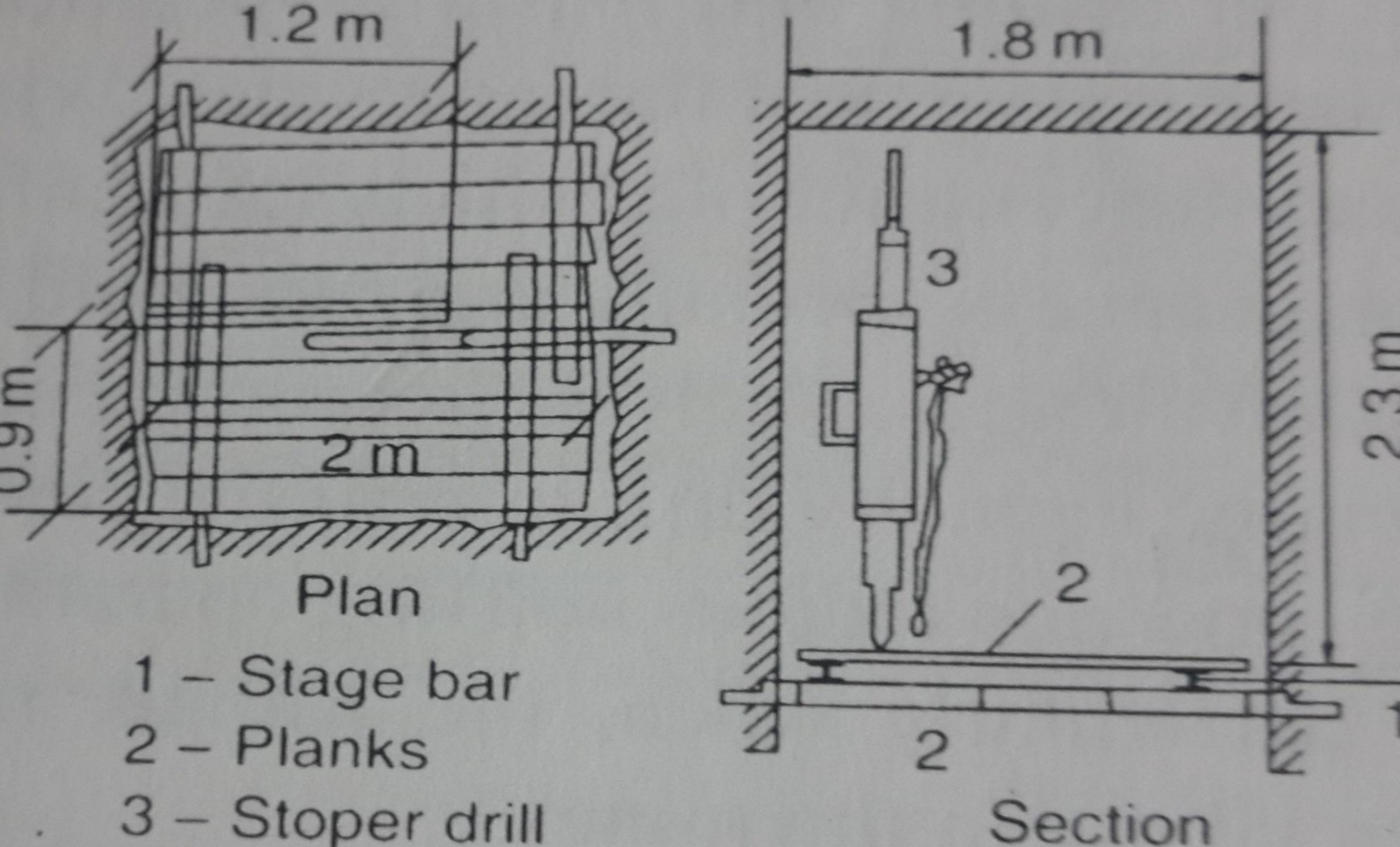
*Raising is very labour intensive.

*High human exposure to risk.

*Raising is slow.

# Open Manual Raise

*Platform, stoper,




1 – Stage bar
2 – Planks
3 – Stoper drill

(a) Open raising

# Reference I

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